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From Tridiagonal Elimination to Symplectic Phase-Space Sweeps:
An Adaptive FrFT–Thomas Framework for Axonal Bundles Under
High-Frequency Electrical Fields

Aman Chawla

Computational Biophysics and Applied Mathematics Group

REAL Institute, Gurugram, India

Email: achawla@alum.mit.edu

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Abstract

The numerical integration of multi-compartmental axonal cable systems classically relies on the Thomas algorithm (tridiagonal Gaussian elimination) running in strictly physical coordinates with $\mathcal{O}(K)$ computational complexity. In this laboratory report, we chronicle the theoretical synthesis of a novel linear-algebraic solver that unifies spatial Gaussian elimination with modal spectral decomposition via continuous phase-space rotations. Beginning with a thought experiment mapping elementary elimination operators across the Fourier bridge, we analyze why naive spectral projections explode into dense $\mathcal{O}(K^2)$ cross-talk.

Motivated by phase-space path integral quantum mechanics, we introduce the Fractional Fourier Transform (FrFT) as a continuous symplectic dial $\alpha(x)$ interpolating between pure coordinate sparsity ($\alpha = 0$) and spectral decoupling ($\alpha = \pi/2$). We evaluate this framework on a high-dimensional distributed architecture (1 CPU per axon) modeling N ephaptically coupled axons. While standard action potentials are purely parabolic and favor pure spatial elimination, driving fields such as high-frequency alternating current (HFAC, 10–50 kHz) nerve blocks introduce pseudo-hyperbolic oscillatory dispersion that breaks classical spatial solvers through phase pollution. We propose a novel, multi-scale, patch-based FrFT–Thomas solver designed to eliminate this phase error on coarse discretizations without incurring global fill-in.

Keywords: Thomas Algorithm, Fractional Fourier Transform, Ephaptic Coupling, Cable Equation, High-Frequency Alternating Current, Microlocal Analysis, Semiclassical Mechanics.

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1 Introduction and Theoretical Motivation

The implicit numerical integration of one-dimensional reaction-diffusion systems—most prominently the Hodgkin-Huxley (HH) and cable equations of computational neuroscience—relies almost universally on the backward Euler or Crank-Nicolson (CN) schemes. When formulated on standard branched or linear neuronal geometries, the resulting linear system at each time step possesses a tridiagonal or tree-like sparse structure. Since the pioneering work of Hines (1984), these systems have been solved using the Thomas algorithm, which is an $\mathcal{O}(K)$ realization of Gaussian elimination without pivoting.

The spatial Thomas algorithm operates strictly in the standard coordinate basis. Decades in parallel, spectral and pseudo-spectral methods have dominated the treatment of homogeneous differential operators by projecting the state vector onto an orthogonal eigenbasis (e.g., via the Discrete Sine or Fourier Transform), decoupling the system into scalar divisions in $\mathcal{O}(K \log K)$ operations.

This laboratory report journals an investigation into the continuum between these two operational paradigms. We address the following foundational and applied questions:

1. What happens algebraically when the elementary steps of local spatial Gaussian elimination are mapped into a global spectral eigenbasis?
2. Can we construct an intermediate operator that bridges spatial sparsity (nearest-neighbor coupling) and spectral decoupling without suffering the computational penalties of both?
3. How does this mathematical duality translate to the biophysics of nerve bundles, particularly when modeling N ephaptically coupled axons driven by high-frequency electric fields?

2 Journal of Conceptual Evolution

2.1 Phase 1: The Spatial vs. Spectral Thought Experiment

We initiated this investigation by asking: *If the Thomas algorithm refuses to cross the Fourier bridge, remaining strictly spatial/local, what occurs if we translate each elementary spatial operation into the Fourier domain, sequence them, and analyze the resulting operator?*

Let $A \in \mathbb{R}^{K \times K}$ be a symmetric, tridiagonal matrix arising from a second-order spatial discretization of $-\partial_{xx}$:

$$A = \begin{pmatrix} b & c & 0 & \dots & 0 \\ a & b & c & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & a & b & c \\ 0 & \dots & 0 & a & b \end{pmatrix} \quad (1)$$

The matrix admits an exact spectral diagonalization $A = F\Lambda F^*$, where F is unitary (e.g., the Discrete Sine Transform, DST-I) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_K)$.

The spatial Thomas algorithm decomposes A into a sequence of $2K - 1$ elementary rank-1 updates. The forward elimination utilizes elementary lower unit shear matrices:

$$L_k = I - m_k e_{k+1} e_k^T, \quad m_k = \frac{a}{b'_k}, \quad k \in \{1, \dots, K-1\} \quad (2)$$

such that $L^{-1} = L_{K-1} L_{K-2} \dots L_1$. Similarly, backward substitution is mediated by diagonal scaling D^{-1} and upper shears $U_k = I - \frac{c}{b'_k} e_k e_{k+1}^T$. In physical space, e_k is a localized coordinate Dirac-like vector; thus, each step requires only $\mathcal{O}(1)$ operations, totaling $\mathcal{O}(K)$.

We then mapped each elementary operator through the similarity transform:

$$\widehat{L}_k = F^* L_k F = I - m_k (F^* e_{k+1}) (e_k^T F) = I - m_k \hat{w}_{k+1} \hat{v}_k^* \quad (3)$$

Because $\hat{w}_{k+1}$ and $\hat{v}_k^*$ are global Fourier modes, every $\widehat{L}_k$ in the spectral domain is a **dense, rank-1 perturbation of the identity**. Stacking them in sequence yields:

$$\widehat{M} = \left(\prod_{k=1}^{K-1} \widehat{U}_k \right) \widehat{D}^{-1} \left(\prod_{k=1}^{K-1} \widehat{L}_k \right) = F^* (U^{-1} L^{-1}) F = F^* A^{-1} F = \Lambda^{-1} \quad (4)$$

Verdict of Phase 1: The stacked operator $\widehat{M}$ analytically solves the original problem, collapsing miraculously into the exact inverse eigenvalue matrix Λ^{-1} . However, applying $2K - 1$ dense rank-1 operators to a vector sequentially costs:

$$\mathcal{C} = (2K - 1) \times \mathcal{O}(K) = \mathcal{O}(K^2) \text{ operations} \quad (5)$$

This demonstrated the *uncertainty principle penalty of linear algebra*: local coordinate operations become non-local frequency mixers. Factoring an operator that is already decoupled in Fourier space via step-by-step shears is an inefficient, $\mathcal{O}(K^2)$ scenic route to compute scalar divisions.

2.2 Phase 2: The Physical Analogy to Quantum Path Integrals

To understand why the algorithm degraded, we mapped the duality onto Feynman's path integral formulations of quantum mechanics. In the short-time propagator $e^{-i\hat{H}\Delta t/\hbar}$, the choice of discretization reflects the operator basis:

- **Configuration Space Path Integral ($\mathcal{D}[x]$):** The potential $\hat{V}$ is diagonal; the kinetic operator $\hat{T} = \hat{p}^2/2m$ is non-diagonal, manifesting as nearest-neighbor kinetic links $(x_j - x_{j-1})^2$. This is the continuous counterpart to tridiagonal matrix coupling (Thomas domain).
- **Momentum Space Path Integral ($\mathcal{D}[p]$):** The kinetic operator is diagonal; the potential operator becomes a dense convolution kernel $\tilde{V}(p - p')$.
- **Phase Space Path Integral ($\mathcal{D}[x]\mathcal{D}[p]$):** Neither x nor p is integrated out. The action is evaluated along trajectories in phase space.

This provided the central insight: *the optimal basis for an operator depends on the localized geometry of its stationary phase in phase space*. If a wavefunction is localized in space (bound state in a deep well), configuration space ($\mathcal{D}[x]$ / Thomas) is optimal. If it is localized in momentum (free scattering wave), momentum space ($\mathcal{D}[p]$ / Spectral) is optimal. If the dynamics oscillate or disperse across both, one must rotate coordinates directly in phase space.

2.3 Phase 3: The Intermediate Transform (FrFT)

Rather than forcing a binary choice between pure space ($\alpha = 0$) and pure frequency ($\alpha = \pi/2$), we introduced the **Fractional Fourier Transform (FrFT)** as a continuous rotation operator in phase space.

Definition 1 (Fractional Fourier Transform). *For an angle $\alpha \in [0, \pi/2]$, the continuous Fractional Fourier Transform $\mathcal{F}_\alpha$ of a function $u(x)$ is defined by the integral kernel:*

$$(\mathcal{F}_\alpha u)(y) = \int_{-\infty}^{\infty} K_\alpha(x, y) u(x) dx \quad (6)$$

where

$$K_\alpha(x, y) = \begin{cases} \sqrt{\frac{1-i\cot\alpha}{2\pi}} \exp\left(i\frac{x^2+y^2}{2}\cot\alpha - i\frac{xy}{\sin\alpha}\right), & \alpha \notin \pi\mathbb{Z} \\ \delta(x-y), & \alpha = 0 \\ \frac{1}{\sqrt{2\pi}}e^{-ixy}, & \alpha = \pi/2 \end{cases} \quad (7)$$

In Wigner phase space (x, k) , the transform $\mathcal{F}_\alpha$ performs a rigid symplectic rotation of the state by angle α :

$$\begin{pmatrix} x_\alpha \\ k_\alpha \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ k \end{pmatrix} \quad (8)$$

By varying α locally, an operator whose classical symbol surface $z = H(x, k)$ has non-zero phase-space curvature can be locally aligned with coordinate axes, minimizing non-local dispersion.

3 Application: Coupled Axonal Bundles under High-Frequency Fields

3.1 Governing Biophysical Equations

Consider a nerve bundle containing N parallel unmyelinated axons running along coordinate $x \in [0, L]$. Each axon $i \in \{1, \dots, N\}$ has radius a_i , intracellular axoplasmic resistivity R_i , and specific membrane capacitance C_m . The axons interact through a shared extracellular resistive medium with resistivity R_e .

The transmembrane potential is $V_{m,i}(x, t) = V_i(x, t) - V_e(x, t)$. The longitudinal cable equation couples to the extracellular field according to:

$$C_m \frac{\partial V_{m,i}}{\partial t} = \frac{a_i}{2R_i} \frac{\partial^2 V_i}{\partial x^2} - I_{\text{ion},i}(V_{m,i}, \mathbf{w}_i) + I_{\text{stim},i}(x, t) \quad (9)$$

where $\mathbf{w}_i = (m_i, h_i, n_i)^T$ represents the Hodgkin-Huxley gating state vectors, and the ionic current is:

$$I_{\text{ion},i} = \bar{g}_{\text{Na}} m_i^3 h_i (V_{m,i} - E_{\text{Na}}) + \bar{g}_{\text{K}} n_i^4 (V_{m,i} - E_{\text{K}}) + g_L (V_{m,i} - E_L) \quad (10)$$

The extracellular potential $V_e(x, t)$ is established by the superposition of transmembrane currents from all N axons (ephaptic coupling) and externally applied fields from stimulation electrodes:

$$V_e(x, t) = V_{e,\text{ext}}(x, t) + R_e \sum_{j=1}^N \pi a_j^2 \left(\frac{a_j}{2R_j} \frac{\partial^2 V_j}{\partial x^2} \right) \quad (11)$$

3.2 Spatiotemporal Discretization and Matrix Topology

Let each axon be discretized along its length into K nodes with uniform spacing Δx . Time is discretized into steps Δt using the Crank-Nicolson formulation. The linearized algebraic system across the entire bundle has dimension $(NK) \times (NK)$.

If ordered such that node k of all N axons is grouped sequentially, the global operator matrix $\mathcal{A}$ is a block-tridiagonal matrix composed of K diagonal blocks and $2(K-1)$ off-diagonal coupling blocks:

$$\mathcal{A} = \begin{pmatrix} \mathbf{B}_1 & \mathbf{C}_1 & & 0 \\ \mathbf{A}_2 & \mathbf{B}_2 & \mathbf{C}_2 & \\ & \ddots & \ddots & \ddots \\ 0 & & \mathbf{A}_K & \mathbf{B}_K \end{pmatrix}, \quad \mathbf{A}_k, \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^{N \times N} \quad (12)$$

In an architecture allocating **1 CPU per axon**, the distributed memory layout prefers an axon-by-axon partitioning:

$$\mathcal{A} = \begin{pmatrix} \mathbf{T}_{11} & \mathbf{E}_{12} & \dots & \mathbf{E}_{1N} \\ \mathbf{E}_{21} & \mathbf{T}_{22} & \dots & \mathbf{E}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{E}_{N1} & \mathbf{E}_{N2} & \dots & \mathbf{T}_{NN} \end{pmatrix} \quad (13)$$

Here, each $\mathbf{T}_{ii} \in \mathbb{R}^{K \times K}$ is a strictly tridiagonal internal diffusion matrix local to CPU i , while each $\mathbf{E}_{ij} \in \mathbb{R}^{K \times K}$ is a **purely diagonal matrix** representing transverse point-to-point ephaptic cross-talk between axon i and axon j .

3.3 Transverse Decoupling via Analytical Eigenbases

Assuming transverse spatial invariance or circular cross-sectional bundle packing, the inter-axon coupling matrix $\mathbf{C}_{\text{trans}} \in \mathbb{R}^{N \times N}$ is symmetric Toeplitz or circulant. Let $\mathbf{Q} \in \mathbb{R}^{N \times N}$ be the orthogonal matrix of discrete Fourier/sine modes across the axon cross-section:

$$\mathbf{Q}^T \mathbf{C}_{\text{trans}} \mathbf{Q} = \mathbf{\Lambda}_{\text{trans}} = \text{diag}(\mu_1, \dots, \mu_N) \quad (14)$$

Applying this unitary rotation across the cluster via global collective communication using `MPI_Alltoall` decouples the bundle into N completely independent longitudinal modes:

$$\tilde{\mathbf{T}}_m \tilde{\mathbf{V}}_m = \tilde{\mathbf{d}}_m, \quad m \in \{1, \dots, N\} \quad (15)$$

where each decoupled modal matrix $\tilde{\mathbf{T}}_m \in \mathbb{R}^{K \times K}$ is tridiagonal:

$$\tilde{\mathbf{T}}_m = \mathbf{T}_{\text{cable}} + \mu_m \mathbf{D}_{\text{ephaptic}} \quad (16)$$

Thus, the problem on each individual CPU reduces to solving a 1D non-homogeneous cable equation.

3.4 Failure of Spatial Discretization Under HFAC Fields

When the nerve bundle is subjected to High-Frequency Alternating Current (HFAC) stimulation (e.g., 10–50 kHz sinusoidal blocking waveforms), the driving source term is:

$$V_{e,\text{ext}}(x, t) = V_0(x) \cos(\omega t), \quad \omega = 2\pi f, \quad f \geq 10 \text{ kHz} \quad (17)$$

Substituting this oscillatory field into the modal cable equation yields an effective temporal-spatial Helmholtz-like equation for the envelope $\hat{u}(x)$:

$$-\lambda^2 \frac{d^2 \hat{u}}{dx^2} + (1 + i\omega\tau) \hat{u} = \hat{S}(x) \quad (18)$$

where $\tau = r_m c_m$ and $\lambda = \sqrt{a/(2R_i g_m)}$. The classical symbol in phase space is:

$$H(x, k) = \lambda^2 k^2 + 1 + i\omega\tau \quad (19)$$

For $\omega\tau \gg 1$, the continuous solutions possess a rapid spatial wavenumber:

$$k_0 \approx \sqrt{\frac{\omega\tau}{2\lambda^2}} \quad (20)$$

The Pollution Effect: Standard second-order finite difference approximations (used in the Thomas algorithm) produce a numerical dispersion relation:

$$k_{\text{num}} = k_0 (1 - \mathcal{O}(k_0^2 \Delta x^2)) \quad (21)$$

Accumulated over an axon of length L , the total phase error scales as $\mathcal{E}_{\text{phase}} \sim \omega^3 L \Delta x^2$. To maintain numerical fidelity under 40 kHz HFAC block, the spatial grid must be refined to $K \geq 10^5$ nodes, rendering implicit simulation computationally prohibitive.

4 The Adaptive FrFT–Thomas Numerical Algorithm

To overcome dispersion errors without global dense matrix inversion, we formulate the **Localized Fractional Phase-Space (FrFT–Thomas) Marching Algorithm**.

4.1 Algorithmic Formulation

The axon is partitioned into M overlapping subdomains (patches) $\Omega_j = [x_{j,\text{start}}, x_{j,\text{end}}]$ of size B nodes, such that $K \approx M \times B$.

1. **Dynamic Phase Angle Assignment:** For each patch j , evaluate the local driving frequency and potential gradient to define the local rotation angle α_j :

$$\alpha_j = \arctan\left(\frac{k_0(x_j)}{1/\lambda}\right) \in \left[0, \frac{\pi}{2}\right) \quad (22)$$

- In resting/unstimulated segments: $\omega \approx 0 \implies \alpha_j = 0$ (Pure Spatial Thomas).
- Under the HFAC electrode: $\omega \gg 1 \implies \alpha_j \rightarrow \pi/2$ (Chirp-Demodulated Frame).

2. **Local Fractional Transform:** Apply the discrete Fractional Fourier Transform $\mathbf{F}_{\alpha_j} \in \mathbb{C}^{B \times B}$ within patch j :

$$\hat{\mathbf{u}}_j = \mathbf{F}_{\alpha_j} \mathbf{u}_j, \quad \hat{\mathbf{T}}_j = \mathbf{F}_{\alpha_j} \tilde{\mathbf{T}}_j \mathbf{F}_{\alpha_j}^* \quad (23)$$

Because α_j matches the local phase-space trajectory, the fast carrier wave $\exp(ik_0 x)$ is mapped to the origin ($k_\alpha \approx 0$). In this rotated frame, $\hat{\mathbf{T}}_j$ is an ultra-narrow band matrix with exponentially decaying off-diagonal entries.

3. **Local Envelope Inversion (Decoupled Solve):** Solve the smooth envelope solution in the rotated patch coordinates:

$$\hat{\mathbf{u}}_j^{(0)} = \left[\text{diag}\left(\hat{\mathbf{T}}_j\right)\right]^{-1} \hat{\mathbf{d}}_j \quad (24)$$

4. **Fractional Patch Interface Coupling (Block-Thomas Riccati Sweep):** Between adjacent patches j and $j+1$, the interface boundary operator is governed by the relative rotation:

$$\mathcal{T}_{j,j+1} = \mathbf{F}_{\alpha_{j+1}} \circ \mathbf{F}_{-\alpha_j} = \mathbf{F}_{\Delta\alpha_j}, \quad \Delta\alpha_j = \alpha_{j+1} - \alpha_j \quad (25)$$

Because $\Delta\alpha_j$ is small, $\mathbf{F}_{\Delta\alpha_j}$ is a tightly banded matrix. The global system across the M patch boundaries is solved using Block-Thomas elimination on the patch impedance matrices $\mathbf{Z}_j$:

$$\mathbf{Z}_{j+1} = \mathcal{T}_{j,j+1} \left(\hat{\mathbf{T}}_j + \mathbf{Z}_j\right)^{-1} \mathcal{T}_{j,j+1}^* \quad (26)$$

5. **Inverse Projection:** Back-substitute across the boundaries and apply the inverse local transform $\mathbf{F}_{-\alpha_j}$ to recover physical transmembrane voltages:

$$\mathbf{V}_{m,j}(x) = \text{Re}\left\{\mathbf{F}_{-\alpha_j} \hat{\mathbf{u}}_j\right\} \quad (27)$$

Table 1: Algorithmic Complexity and Resource Comparison for Solving One Longitudinal Axon System (K Nodes, M Patches of Size B).

Metric	Standard Thomas	Pure Spectral (DST)	Adaptive FrFT–Thomas
Mathematical Basis	Coordinate (e_k)	Eigenmodes (v_k)	Local Phase Manifold (x, k)
Rotation Angle α	0	$\pi/2$	Adaptive $\alpha(x) \in [0, \pi/2)$
FLOP Complexity	$\mathcal{O}(K)$ ($\sim 8K$)	$\mathcal{O}(K \log K)$	$\mathcal{O}(K \log B + MB_{\text{band}})$
Arithmetic Type	Pure Real	Real / Complex	Adaptive Real/Complex
Phase Pollution ($\omega \gg 1$)	Severe ($\mathcal{O}(\omega^3 \Delta x^2)$)	Zero (for const. coeff.)	Minimized via Demodulation
Parallel Partitioning	Inhibited (Sequential)	Global Transpose	Natural Substructuring

4.2 Theoretical Complexity and Flop Comparison

5 Proposed Research and Implementation Program

To validate the theoretical architecture developed in this report, we propose a four-phase research program.

5.1 Phase I: Operator Sparsity Quantification (Weeks 1–4)

- Implement the discrete Fractional Fourier Transform via the fast Ozaktas chirp-z algorithm.
- Compute the off-diagonal Frobenius decay rate of $\mathbf{F}_\alpha \mathbf{A} \mathbf{F}_\alpha^*$ as a function of $|\alpha - \alpha_{\text{opt}}|$ for differential operators with variable coefficients $V(x)$.
- Establish the maximum patch overlap width B_{band} required to truncate the interface operator $\mathcal{T}_{j,j+1}$ below machine precision $\epsilon_{\text{mach}} = 10^{-16}$.

5.2 Phase II: The Monolithic 1D Axon Benchmark under HFAC (Weeks 5–8)

- Construct a numerical benchmark modeling a single unmyelinated squid giant axon ($L = 50$ mm) subjected to a 30 kHz sinusoidal blocking electrode at $x \in [20, 25]$ mm.
- Evaluate the numerical dispersion error and action potential block threshold comparing:
 1. Standard Crank-Nicolson with fine spatial mesh ($K = 10^5$).
 2. Standard Crank-Nicolson with coarse spatial mesh ($K = 10^3$).
 3. Adaptive FrFT–Thomas solver with coarse spatial mesh ($K = 10^3, M = 20, B = 50$).
- Measure wall-clock execution time and memory bandwidth saturation on modern AVX-512 CPU architectures.

5.3 Phase III: Distributed Ephaptic Bundle Implementation (Weeks 9–14)

- Write an MPI/C++ simulation harness allocating 1 CPU core per axon for $N = 1024$ axons.
- Implement the transverse decoupling step using parallel FFT libraries (FFTW / Intel MKL) executing inter-node `MPI_Alltoallv` transforms.
- Quantify parallel scaling efficiency, network latency overhead, and load-balance across non-homogeneous axon diameters.

5.4 Phase IV: Non-Homogeneous and Inelastic Extensions (Weeks 15–20)

- Extend the formulation to myelinated fibers (bifurcated Ranvier nodes where $V(x)$ has step-discontinuities).
- Incorporate 3D non-homogeneous extracellular conductivities by linking the transverse eigen-solver to a boundary element method (BEM) representation of the nerve cuff geometry.
- Target peer-reviewed publication in the *Journal of Computational Physics* or *IEEE Transactions on Biomedical Engineering*.

6 Conclusion

This report has demonstrated that the dichotomy between the spatial Thomas algorithm and spectral eigenbasis methods is an artifact of restricting linear algebra to rigid coordinate projections. Through the lens of semiclassical mechanics and the Fractional Fourier Transform, Gaussian elimination can be continuously steered across phase space.

While pure action potential diffusion operates optimally at $\alpha = 0$, high-frequency electrical neuro-stimulation drives the system into an oscillatory regime where spatial elimination breaks down. The proposed Adaptive FrFT–Thomas solver reconciles this conflict, creating an efficient, multi-scale computational framework for next-generation bioelectronic modeling.

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